SENTHIL PUBLIC SCHOOL, DHARMAPURI

AFFILIATED TO CBSE, NEW DELHI (AFFL. NO. 1930248)

CLASS: CLASS 8 (CBSE)	**SUBJECT: MATHEMATICS & PHYSICAL SCIENCE**	**DATE: 18-OCT-2026**
TIME: 45 MINS	**PHASE: Phase 2: 3-Day Checkpoint (Days 1–3 Audit | Day 4 Morning)**	**MAX. MARKS: 25**

[PART 1: MATHEMATICS (13 MARKS)]

--- PART 1: MATHEMATICS (13 MARKS) — SECTION A: OBJECTIVE & FORMULAS ---

Q.1 [MTH-P2-Q01] Which of the following algebraic identities is universally correct for all real values of a and b? **[1]**
(a) $(a + b)^2 = a^2 + b^2$ (b) $(a - b)^2 = a^2 - 2ab + b^2$ (c) $(a + b)(a - b) = a^2 + b^2$ (d) $(a + b)^2 = a^2 - 2ab + b^2$

Q.2 [MTH-P2-Q02] The coefficient of x^2 in the expanded form of $(3x - 4)^2$ is: **[1]**
(a) 9 (b) -24 (c) 16 (d) -9

Q.3 [MTH-P2-Q03] The value of $(-2/3)^4$ expressed as a positive rational number is: **[1]**
(a) 16/81 (b) -16/81 (c) 8/27 (d) -8/27

--- PART 1: MATHEMATICS — SECTION B: VERY SHORT ANSWER ---

Q.4 [MTH-P2-Q04] Factorise the algebraic expression completely by grouping terms: 15xy - 6x + 5y - 2. **[2]**

--- PART 1: MATHEMATICS — SECTION C: SHORT ANSWER & NUMERICAL ---

Q.5 [MTH-P2-Q05] A cuboid has dimensions 60 cm x 54 cm x 30 cm. How many small cubes with side 6 cm can be placed in the given cuboid? **[3]**

--- PART 1: MATHEMATICS — SECTION D: LONG ANSWER & DERIVATION ---

Q.6 [MTH-P2-Q06] (a) A closed cylindrical storage tank of radius 7 m and height 3 m is made from a sheet of metal. How much sheet of metal is required? (Take $\pi = 22/7$). (b) If the metal costs ☐ 45 per square meter, calculate the total cost. **[5]**

[PART 2: PHYSICAL SCIENCE (12 MARKS)]

--- PART 2: PHYSICAL SCIENCE (12 MARKS) — SECTION A: OBJECTIVE & ASSERTION-REASON ---

Q.7 [SCI-P2-Q07] Pressure is defined as the force acting per unit area of a surface. Its standard S.I. unit is: **[1]**
(a) Newton (N) (b) Joule (J) (c) Pascal (Pa) (d) Watt (W)

Q.8 [SCI-P2-Q08] ASSERTION (A): The shoulder strap of a school bag is made broad. REASON (R): A broad strap increases the surface area of contact, thereby reducing the pressure exerted on the shoulder. **[1]**
(a) Both (A) and (R) are true and (R) is correct explanation. (b) Both (A) and (R) are true but (R) is NOT correct explanation. (c) (A) is true, (R) is false. (d) (A) is false, (R) is true.

--- PART 2: PHYSICAL SCIENCE — SECTION B: VERY SHORT CONCEPT DRILL ---

Q.9 [SCI-P2-Q09] Why is rolling friction substantially less than sliding friction? Give one practical engineering application where sliding friction is replaced by rolling friction. **[2]**

--- PART 2: PHYSICAL SCIENCE — SECTION C: SHORT ANSWER & APPLICATION ---

Q.10 [SCI-P2-Q10] A wooden block of dimensions 20 cm x 10 cm x 5 cm of mass 4 kg is kept on a tabletop. Calculate the maximum pressure exerted by the block on the tabletop in Pascals. (Take g = 10 m/s²). **[3]**

--- PART 2: PHYSICAL SCIENCE — SECTION D: LONG ANSWER & DIAGRAM TASK ---

Q.11 [SCI-P2-Q11] (a) State the two Laws of Reflection of Light. (b) Draw a neat labeled ray diagram showing the incident ray, reflected ray, normal line, angle of incidence (i) and angle of reflection (r). **[5]**